



ReDI School of
Digital Integration

Data Circle 2025 Sprint 1

Project A: Flu Shots



We use tech to connect human potential and
opportunity with dignity & humility

**Team "Halva" - Project A-
Flu Shot Learning: Predict H1N1 and Seasonal
Flu Vaccines**

Project Goal

Predict how likely individuals are to receive their H1N1 and seasonal flu vaccines.

Specifically, predicting two probabilities: one for **h1n1_vaccine** and one for **seasonal_vaccine**.

Each row in the dataset represents one person from USA who responded to the National 2009 H1N1 Flu Survey.

Objective

- ❑ Exploratory analysis of the data
- ❑ Cleaning and preparing the data for training

Methods

- ❑ Data cleaning
- ❑ Data transformation
- ❑ Exploratory data analysis

Data

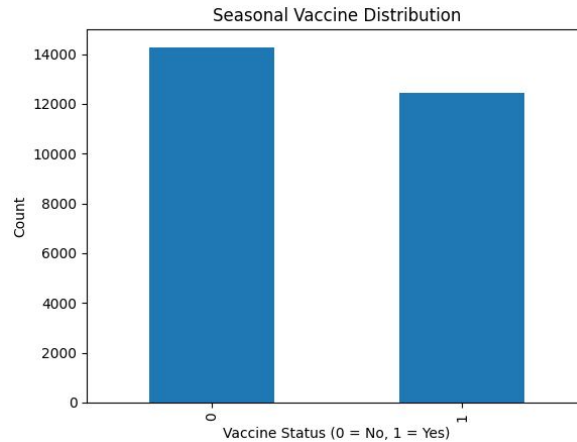
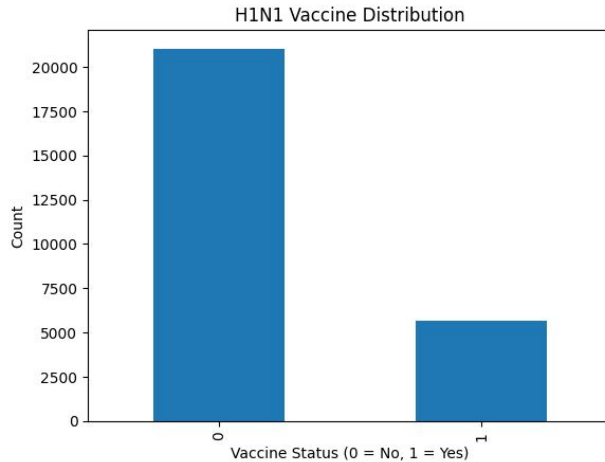
- ❑ 26707 rows - 38 Feature
- ❑ Target variables: h1n1_vaccine, seasonal_vaccine
- ❑ Behavioral features
- ❑ Opinion features
- ❑ Demographic features

Data Cleaning

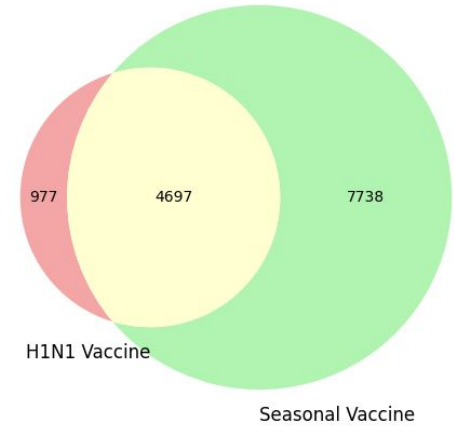
- ❑ Filling missing Data
- ❑ Deciding whether to drop seemingly unnecessary columns
- ❑ Feature encoding

Main Findings

Target variable Distribution



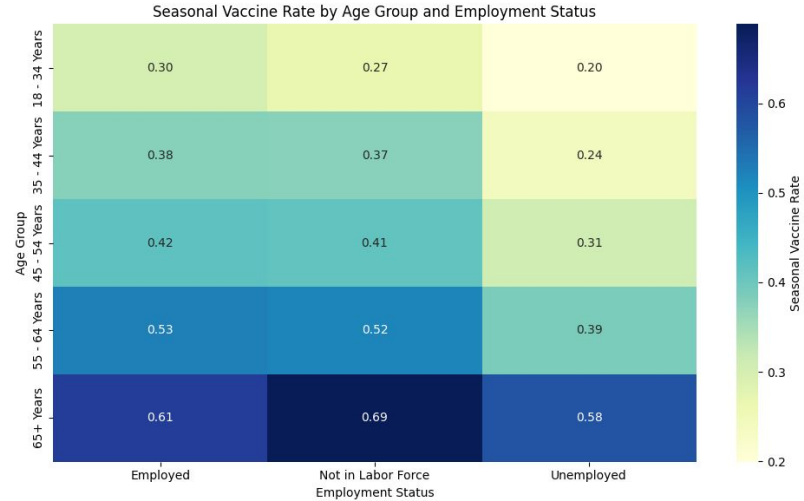
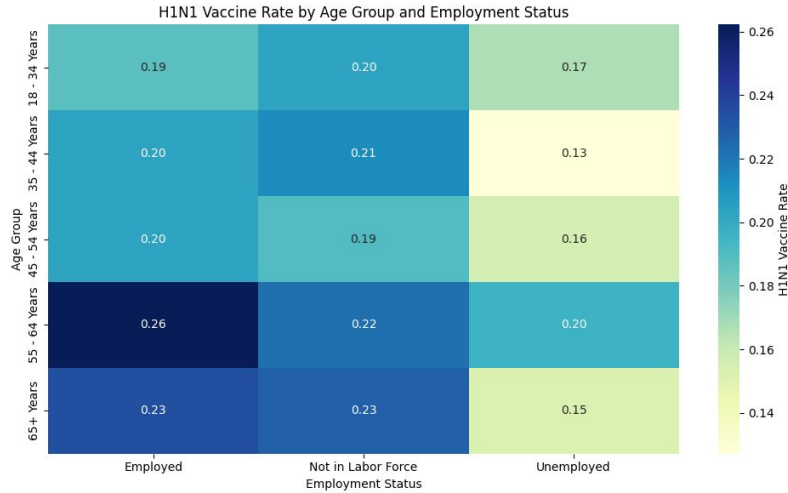
Overlap Between H1N1 and Seasonal Vaccinations



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Main Findings

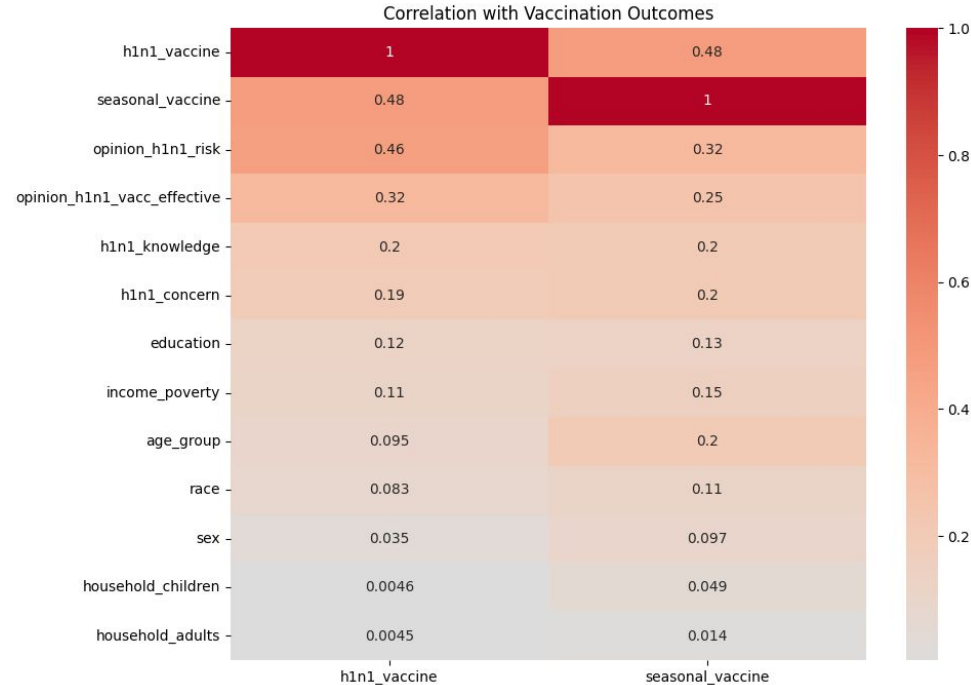
Demographic Influence



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Main Findings

Feature Correlation



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Challenges

- Difficulties setting up and syncing GitHub
- Choosing the correct encoding strategies (label encoder, ordinal encoder)
- Plotting encoded features

Conclusion

- ❑ We have a good understanding of the data and the correlation of the features.
- ❑ The Data is now clean with no missing values.
- ❑ The Data is now encoded and ready for training.

Thank you